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Not all breakpoints are created equally: test-retest reliability and bilateral differences of SmO₂ breakpoints during continuous cycling step tests

Authors:

Philip Skotzke^{1,2}, Massimo Sacchetti¹, Tim Meyer²

¹ Department of Movement, Human and Health Sciences, University of Rome “Foro Italico”, Rome, Italy

² Institute of Sport and Preventive Medicine, University of Saarland, Campus B8.2, 66123 Saarbrücken, Germany

Numerous methods exist to identify muscle oxygen saturation (SmO₂) breakpoints (BPs) to complement whole-body exercise thresholds [1], but their test-retest reliability remains largely unexplored. This exploratory analysis (a) compares test-retest reliability of different BP detection methods and (b) examines whether the power output at the BPs differs between the dominant (DOM) and non-dominant (NDOM) leg and averaged SmO₂ (AVG). We hypothesized that reliability would differ meaningfully across methods, but no differences exist between sides.

A data set of 36 (n = 12 males, 3 tests each) continuous cycling step tests (start 1 W/kg, increase 0.5 W/kg every 5 min) was analysed with SmO₂ measured at the vastus lateralis. Twelve methods were applied across BP1 (8 variants) and BP2 (10 variants): visual identification; double-segmented regression on raw SmO₂, stage-mean SmO₂, stage slope, and the power/SmO₂ ratio (both BPs); log-log regression on stage-mean SmO₂ and the power/SmO₂ ratio (BP1 only); Dmax and modDmax on stage-mean SmO₂ and the power/SmO₂ ratio (BP2 only); and the inflections of the 2nd derivative of a 4th-degree polynomial (both BPs).

The SmO₂ slope and poly methods failed to identify the BP in more than 33% of cases and are excluded. Thus, 15 of 18 method×BP variants are reported. The ICC(2,1) ranged from 0.25 – **0.88**, CV from 3.3 – 22.0 % and SEM from 9 – 51 W. Pairwise bootstrapped comparisons revealed significant differences in test-retest reliability between ~17 % of methods when expressed as within-subject SD and ~19 % when expressed as CV (p < 0.05). ~8 % of the comparisons between DOM, NDOM and AVG yielded significantly different power outputs (p < 0.05). The log-log method on the power/SmO₂ ratio of AVG was most reliable for BP1 (ICC 0.81, CV 8.3 %, SEM 17 W) and the **Dmax method on the power/SmO₂ ratio most reliable for BP2 (ICC 0.91, CV 4.2 %, SEM 9 W).**

Test-retest reliability varied substantially between BP methods, with the best-performing methods not reaching typical reliability of whole-body thresholds. Visual identification may be the most reliable, but precision is limited by the protocol. Practitioners need to understand the uncertainty of methods when interpreting longitudinal data from NIRS. No clear systematic side differences emerged, suggesting single-probe measurement may suffice.

References

1. Sendra-Pérez, C. et al. Sci Rep. 2023;13.